Binary Arithmetic

Computer Organization

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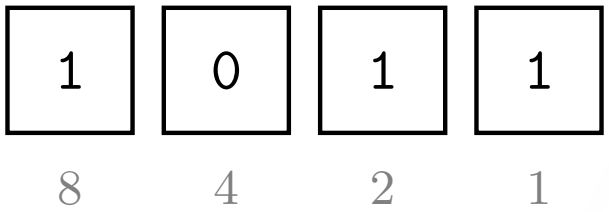
number representation

a rule for encoding numbers

as patterns of bits

unsigned binary

Each bit stands for a power of two. The value of a pattern is the sum of the weights whose bit is 1.



$$8 + 2 + 1 = 11$$

With n bits we can represent every whole number from 0 to $2^n - 1$.

To go the other way, we repeatedly divide by two. Each remainder is one bit, read from the last up to the first.

number	halved	remainder
11	5	1
5	2	1
2	1	0
1	0	1

reading the remainders upward gives 1011

hexadecimal

Long bit patterns are hard to read. **Hexadecimal** replaces each group of four bits with one symbol.

bits	hex	bits	hex
0000	0	1000	8
0001	1	1001	9
0010	2	1010	A
0011	3	1011	B
0100	4	1100	C
0101	5	1101	D
0110	6	1110	E
0111	7	1111	F

We split the bits into groups of four and write the symbol for each group.

1011 0110

becomes

B6

signed numbers

With fixed-length numbers, a negative value has to be represented by a *positive* bit pattern.

Given a number n, we define its negation $-n$ as any number satisfying

$$n + (-n) = 0.$$

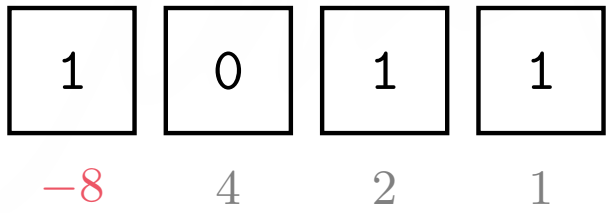
To negate a decimal number, we take two steps.

49	the value 49
50	subtract each digit from 9
51	add one, giving a representation of -49

To negate a binary number we take two similar steps.

0101	the value 5
1010	subtract each bit from 1 (invert each bit)
1011	add one, giving a representation of -5

This is called **twos-complement**.



Adding one to all-ones rolls over to zero, so all-ones must be -1.

$$\begin{array}{r} 1111 \\ +\ 0001 \\ \hline 1\ 0000 \end{array}$$

With n bits, two's complement covers a balanced range around zero.

$$-2^{n-1} \text{ to } 2^{n-1} - 1$$

four bits run from -8 to 7

An older scheme, **sign and magnitude**, uses the top bit only as a sign and the rest as the size.

0110 is 6 1110 is -6

it has two zeros and needs more complex arithmetic, so two's complement is preferred